



B&R CompactFlash

FAQ Changing Over from Silicon Systems to B&R CF

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I Version information

Version	Date	Comment	Author
1.0	25.03.2009	First edition	ROG
1.10	03.11.2009	Various expansions and 16GB CF	ROG

Table 1: Versions

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1 Introduction

The following document provides information that must be taken into consideration when changing over from Silicon Systems CFs to B&R CFs.

Information

Information in this document is based on the actual situation at the time this document was created and should serve as a guide for changing over from Silicon Systems CFs to B&R CFs. For the latest information, make sure to always check the documentation for the respective product.

2 Products affected

Silicon Systems CFs	B&R CFs
5CFCRD.0512-03 CompactFlash 512 MB SSI	5CFCRD.0512-04 CompactFlash 512 MB B&R
5CFCRD.1024-03 CompactFlash 1024 MB SSI	5CFCRD.1024-04 CompactFlash 1024 MB B&R
5CFCRD.2048-03 CompactFlash 2048 MB SSI	5CFCRD.2048-04 CompactFlash 2048 MB B&R
5CFCRD.4096-03 CompactFlash 4096 MB SSI	5CFCRD.4096-04 CompactFlash 4096 MB B&R
5CFCRD.8192-03 CompactFlash 8192 MB SSI	5CFCRD.8192-04 CompactFlash 8192 MB B&R
	5CFCRD.016G-04 CompactFlash 16GB B&R

Table 2: Comparison of Silicon Systems and B&R CFs

The B&R CFs 5CFCRD.XXXX-04 are available in sizes from 512 MB to 16GB.

3 Important modifications

3.1 UDMA mode

B&R CFs 5CFCRD.XXXX-04 use UDMA (1-4) to support an additional IDE mode, which was not yet supported by Silicon Systems CFs.

This mode increases speed significantly, especially for sequential data transfer.

UDMA CFs can run on the following operating systems without any known problems:

- MS-DOS
- Windows XP
- Windows XP Embedded
- Windows CE 6.0

Information:

B&R CFs are not supported by Windows CE <= 5.0

Information:

In VxWorks, problems occur if the BIOS for a UDMA CF starts in UDMA mode.

In this case, VxWorks will not boot without modification of the boot loader!

On B&R devices, this was observed on PP300/400 and PPC300 devices.

In order to support UDMA mode, it's necessary to use a new boot loader that supports UDMA in VxWORKS and "Automation Runtime".

If this is not possible, UDMA mode can be deactivated in BIOS as a work-around.

This work-around also has to be used for operating systems that do not support UDMA.

3.1.1 Operation with Automation Runtime

The new UDMA boot loader is supported starting with PVI Transfer >= V3.2.3.8, which is available on the B&R homepage as PVI Development Setup >= V2.06.00.3011!

Information:

The 16GB CF 5CFCRD.016G-04 is not currently supported by PVI Transfer!

3.2 Disk geometry changed

Important:

B&R CFs have a different disk geometry than Silicon Systems CFs.

Silicon Systems CF	Capacity (bytes)	Number of sectors	Number of cylinders	Number of heads	Number of sectors/tracks
5CFCRD.0512-03	521,773,056	1,019,088	1011	16	63
5CFCRD.1024-03	1,047,674,880	2,046,240	2030	16	63
5CFCRD.2048-03	2,098,446,336	4,098,528	4066	16	63
5CFCRD.4096-03	4,224,761,856	8,251,488	8166	16	63
5CFCRD.8192-03	8,455,200,768	16,514,064	16,383	16	63

Table 3: Disk geometry of Silicon Systems CFs

B&R CF	Capacity (bytes)	Number of sectors	Number of cylinders	Number of heads	Number of sectors/tracks
5CFCRD.0512-04	512,493,328	1,000,944	993	16	63
5CFCRD.1024-04	1,024,966,656	2,001,888	1986	16	63
5CFCRD.2048-04	2,048,901,120	4,001,760	3970	16	63
5CFCRD.4096-04	4,110,188,544	8,027,712	7964	16	63
5CFCRD.8192-04	8,195,604,480	16,007,040	15,580	16	63
5CFCRD.016G-04	16,239,476,736	31,717,728	16,383 ⁽¹⁾	16	63

Table 4: Disk geometry of B&R CFs (1) CHS addressing is limited to 8GB. Larger drives should be operated in LBA mode

In these two tables, you can see that B&R CFs have a slightly lower capacity than Silicon Systems CFs.

Information:

Before each changeover, make sure that the existing application will fit on the B&R CF with the same size.

Additionally, there are various limitations when restoring Silicon Systems images on a B&R CF, which are described in the following section.

4 Restoring CF images (Silicon Systems to B&R CFs)

4.1 Commercial disk image products

No problems were found during various tests when disk image products that are available commercially such as Norton Ghost, Powerquest Drive Image, etc. are used to restore images created on a Silicon Systems CF to a B&R CF.

However, experience from previous CF changeovers shows that problems can occur in rare cases, especially when changing partition sizes. (System doesn't boot)

This problem occurs right away during the first boot procedure and is therefore simple to test!

The error usually occurs in relation with boot loaders if they are located at the wrong position or are fragmented because of the partition size modification.

Important:

B&R recommends that you do not change the size of the boot partition when restoring images.

In some cases, restoring images on a data carrier with a different type/manufacturer can also cause problems with commercial disk image products.

Therefore, the boot procedure for a restored image should always be tested after switching over from a Silicon Systems CF to a B&R CF.

Information:

For more detailed information and limitations regarding the cloning of data carriers with commercial disk image products, please refer to the documentation of the respective disk image product.

4.2 B&R Embedded OS Installer

Important:

B&R CFs 5CFCRD.XXXX-04 up to 8GB are supported starting with Version >= V3.10 of the B&R Embedded OS Installer.

The 5CFCRD.016G-04 with 16GB is not currently supported by the Embedded OS installer!

B&R CFs 5CFCRD.XXXX-04 of the same size have slightly less capacity than Silicon Systems CFs 5CFCRD.XXXX-03, therefore images created from Silicon Systems CFs using "Create Image" cannot be restored on B&R CFs of the same size!

If FAT16/FAT32 is used as the file system, a Silicon Systems image can be restored on a B&R CF of the next larger size.

In general, NTFS images or images with a file system other than FAT16/FAT32 from Silicon Systems CFs cannot be restored on a B&R CF.

Information:

For detailed information and limitations regarding the cloning of CFs with the B&R Embedded OS installer, please refer to the online Help for the B&R Embedded OS installer.

4.3 PVI Transfer

Important:

B&R CFs 5CFCRD.XXXX-04 are supported starting with PVI Transfer >= V3.2.3.8, which is available on the B&R homepage as PVI Development Setup >= V2.06.00.3011!
The 5CFCRD.016G-04 with 16GB is not currently supported by PVI Transfer!

Information:

The new UDMA boot loader is supported starting with PVI Transfer >= V3.2.3.8, which is available on the B&R homepage as PVI Development Setup >= V2.06.00.3011!

B&R CFs 5CFCRD.XXXX-04 of the same size have slightly less memory than Silicon Systems CFs 5CFCRD.XXXX-03, therefore images created from Silicon Systems CFs using "Create image file from CF" cannot be restored on B&R CFs of the same size!

If an Automation Runtime system is located on the Silicon Systems source CF, then a Silicon Systems image can be restored on a B&R CF with the next size up.

Information:

For detailed information and limitations regarding the cloning of CFs with the PVI transfer tool, please refer to the online Help for the PVI transfer tool.

4.4 Plug and play mechanism in operating systems

When using operating systems such as Windows XP and Windows XP Embedded, a new data carrier is identified by the Plug and Play mechanism.

The new data carrier is located just once during the initial boot procedure after the image has been restored and then functions without any problems.

If a Windows XP Embedded operating system is booted with enabled write filter, then the new data carrier is detected during each boot procedure until it has been booted at least once with disabled write filter.

Important:

If manual device changes were made, e.g. switching off the disk cache for the Silicon Systems CF, then these settings are lost and must be set again manually for the B&R CF.

In the event that other Plug and Play operating systems also detect a new data carrier, similar behavior must be expected.

If this is the case, then you also have to check whether manual changes must be made or not!

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